

Homework #2: Interval mapping

Due 9 Mar

Grab the comma-delimited data file at

<http://www.biostat.wisc.edu/~kbroman/teaching/qtltopics/hw2.csv>

This is simulated data from a backcross with 300 individuals, with a single quantitative phenotype. A selective genotyping strategy was used.

1. Use each of standard interval mapping and Haley-Knott regression to map QTL in this cross. Also, use a permutation test to establish significance of identified QTL, and calculate 1.5-LOD support intervals for the locations of inferred QTL.

Do two versions of the permutation test: the usual kind plus a stratified permutation test (with individuals stratified by the amount of genotyping).

2. How do the results change if you omit the individuals that were not genotyped?

To drop the non-genotyped, individuals, use code like

```
subcross <- subset(mycross, ind=(ntyped(mycross) > 0))
```

3. How do the results change if you consider only the genotyped individuals and define a binary trait (high/low phenotype)?

To define the binary trait, use code like

```
phe <- subcross$pheno[,1]
phe <- as.numeric(phe > mean(phe))
subcross$pheno[,1] <- phe
```